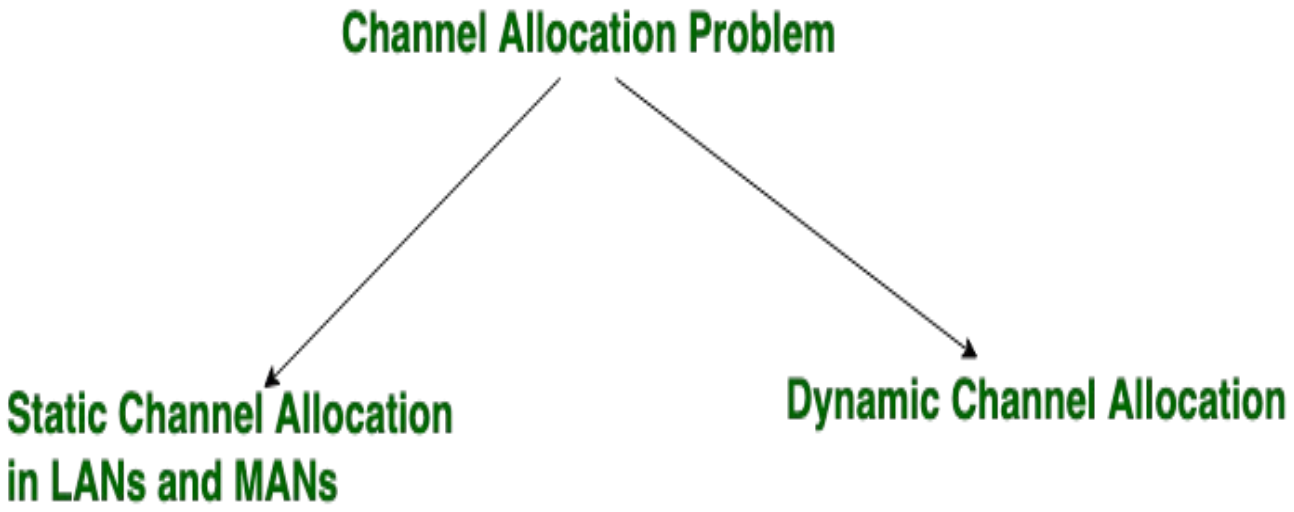
Unit 3
Part-2

Medium Access Control Sub Layer:

- Channel Allocation Problem:
- The Channel Allocation Problem arises in communication networks when multiple devices need to share a limited number of communication channels. The goal is to efficiently allocate channels to devices while avoiding interference, reducing congestion, and optimizing network performance. Channel allocation is a process in which a single channel is divided and allotted to multiple users to carry user specific tasks. The user's quantity may vary every time the process takes place.
- If there are N number of users and the channel is divided into N equal-sized sub-channels then each user is assigned one portion. If the number of users is small and doesn’t vary at times, then Frequency Division Multiplexing can be used as it is a simple and efficient channel bandwidth-allocating technique.

- Channel Allocation Schemes
- Channel allocation problem can be solved by two schemes: Static Channel Allocation in LANs and MANs, and Dynamic Channel Allocation.



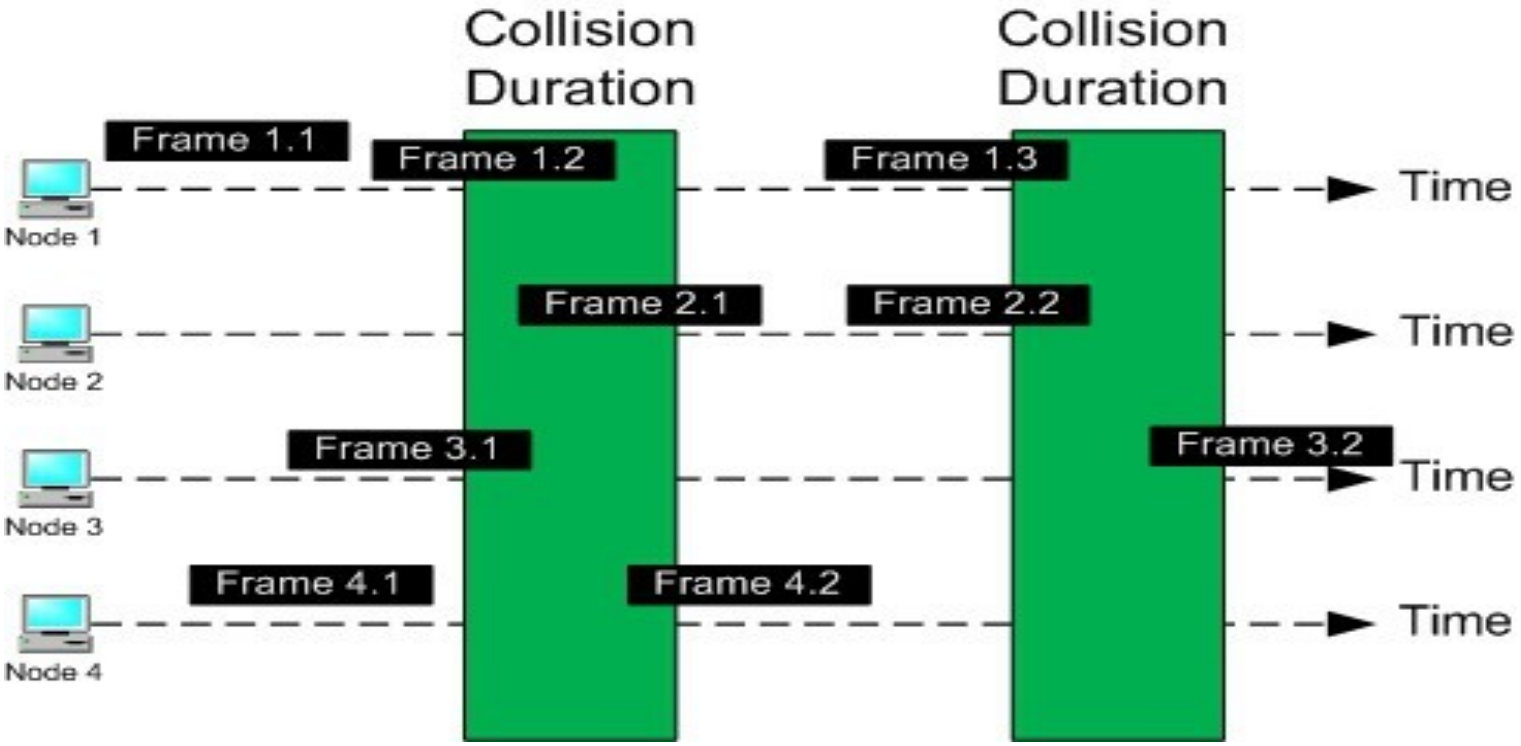
- Static Channel Allocation in LANs and MANs
- It is the classical or traditional approach of allocating a single channel among multiple competing users using Frequency Division Multiplexing (FDM). if there are N users, the frequency channel is divided into N equal-sized portions (bandwidth), and each user is assigned one portion. since each user has a private frequency band, there is no interference between users.

- Dynamic Channel Allocation
- In dynamic channel allocation scheme, frequency bands are not permanently assigned to the users. Instead channels are allotted to users dynamically as needed, from a central pool. The allocation is done considering a number of parameters so that transmission interference is minimized.
- This allocation scheme optimises bandwidth usage and results is faster transmissions.
- Dynamic channel allocation is further divided into:
- Centralised Allocation
- Distributed Allocation

- Some of the possible assumptions include:
- Station Model: Assumes that each of N stations independently produce frames. The probability of producing a packet in the interval IDt where I is the constant arrival rate of new frames.
- Single Channel Assumption: In this allocation all stations are equivalent and can send and receive on that channel.
- Collision Assumption: If two frames overlap in time-wise, then that's collision. Any collision is an error, and both frames must re transmitted. Collisions are only possible error.
- Time can be divided into Slotted or Continuous and Stations can sense a channel is busy before they try it.

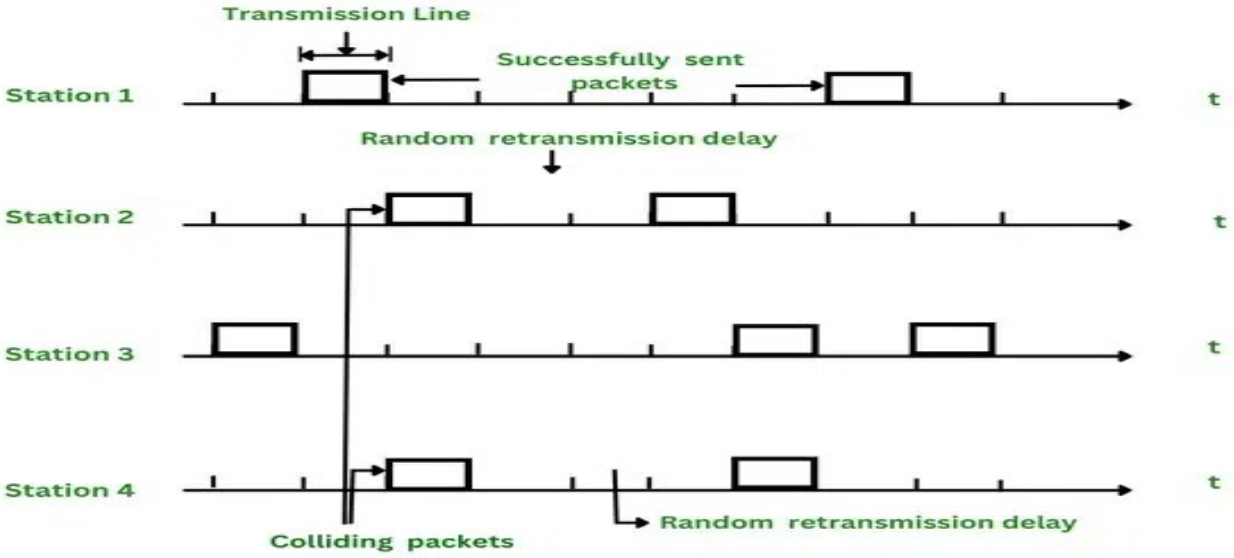
- ALOHA:
- Aloha is a type of Random access protocol it was developed at the University of Hawaii in early 1970, it is a LAN-based protocol this type there are more chances of occurrence of collisions during the transmission of data from any source to the destination, Aloha has two types one Pure Aloha and another Slotted Aloha.

- Pure Aloha
- Pure Aloha can be termed as the main Aloha or the original Aloha. Whenever any frame is available, each station sends it, and due to the presence of only one channel for communication, it can lead to the chance of collision.
- In the case of the pure aloha, the user transmits the frame and waits till the receiver acknowledges it, if the receiver does not send the acknowledgment, the sender will assume that it has not been received and sender resends the acknowledgment.


Collision Duration
Collision Duration
Frame 1.1
Frame 1.2
Frame 1.3
Time
Node 1
Frame 2.1
Frame 2.2
Time
Node 2
Frame 3.1
Frame 3.2
Time
Node 3
Frame 4.1
Frame 4.2
Time
Node 4

- The first step is for each node in the network to listen to the communication channel and wait for an idle period. If the channel is busy, the node waits for the current transmission to finish before proceeding
- Once the node detects an idle period, it transmits its data packet onto the communication channel
- After transmitting the packet, the node waits a random amount of time before checking if the transmission was successful
- The node waits for another random amount of time before retransmitting the packet when the node detects a collision
- If the transmission was successful (i.e., no collisions), the node can then listen for an acknowledgement (ACK) signal from the receiver to confirm that the packet was received successfully. If the node does not receive an ACK signal, it assumes the transmission failed. Then, it will retransmit the packet after waiting for another random amount of time
- The node will continue to retransmit the packet until a successful transmission is achieved or a maximum number of retries is reached
- Once the node has successfully transmitted its packet and received an ACK signal, it can then begin the process of listening for another idle period and transmitting another packet

- Slotted Aloha
- Slotted Aloha is simply an advanced version of pure Aloha that helps in improving the communication network. A station is required to wait for the beginning of the next slot to transmit. The vulnerable period is halved as opposed to Pure Aloha.
- Slotted Aloha helps in reducing the number of collisions by properly utilizing the channel and this basically results in the somehow delay of the users. In Slotted Aloha, the channel time is separated into particular time slots.


Transmission Line
Successfully sent packets
Station 1
t
Random retransmission delay
Station 2
t
Station 3
t
Station 4
t
Random retransmission delay
Colliding packets

Pure Aloha	Slotted Aloha
In this Aloha, any station can transmit the data at any time.	In this, any station can transmit the data at the beginning of any time slot.
In this, The time is continuous and not globally synchronized.	In this, The time is discrete and globally synchronized.
Vulnerable time for Pure Aloha = 2 x Tt	Vulnerable time for Slotted Aloha = Tt
In Pure Aloha, the Probability of successful transmission of the data packet $= G \times e^{-2G}$	In Slotted Aloha, the Probability of successful transmission of the data packet $= G \times e^{-G}$
In Pure Aloha, Maximum efficiency = 18.4%	In Slotted Aloha, Maximum efficiency = 36.8%
Pure Aloha doesn't reduce the number of collisions to half.	Slotted Aloha reduces the number of collisions to half and doubles the efficiency of Pure Aloha.

What is Collision?

- In computer networking (CN), a collision is when two or more devices transmit data at the exact same time on a shared network medium, causing the data packets to interfere and become corrupted. To handle this, protocols like Carrier Sense Multiple Access with Collision Detection (CSMA/CD) are used, where devices listen before transmitting. If a collision is detected, all devices involved will stop transmitting, wait for a random amount of time, and then retransmit their data.

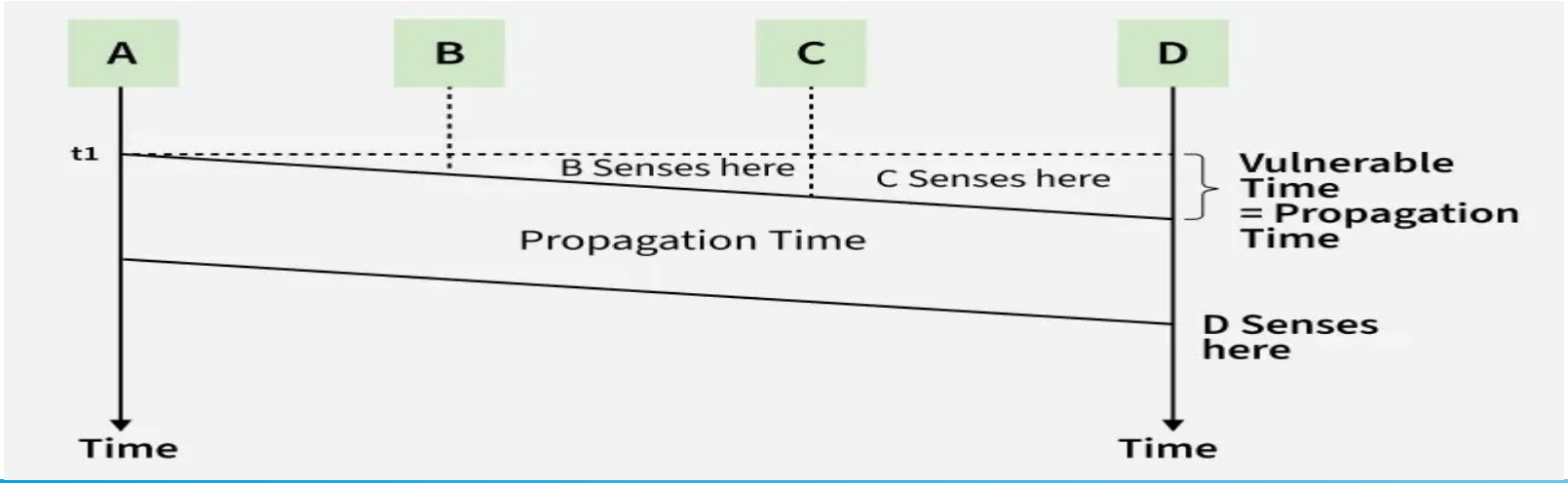
Carrier Sense Multiple Access (CSMA)

- Carrier Sense Multiple Access (CSMA) is a method used in computer networks to help devices share a communication channel without interfering with each other. Before a device sends data, it listens to the channel (or senses the carrier) to check if it’s free.
- If the channel is busy, the device waits until it becomes idle.
- This helps prevent data collisions, which can happen when two devices try to send data at the same time.
- CSMA is widely used in networking technologies like Ethernet and Wi-Fi.

- Types of CSMA Protocol
- There are two main types of Carrier Sense Multiple Access (CSMA) protocols, each designed to handle how devices manage potential data collisions on a shared communication channel. These types differ based on how they respond to the detection of a busy network:

  1.CSMA/CA

  2.CSMA/CD

- Vulnerable Time in CSMA
- Vulnerable time is the short window in which there's a risk of collision between two devices trying to send data on the network at the same time. In CSMA-based networks like Ethernet or Wi-Fi, each device listens to the channel before transmitting.
- 

- If the channel seems free, the device waits a tiny moment and then begins sending.
- However, another device might also sense the channel as free at the same time and they both may start transmitting causing a collision.
- This small time window, where such collisions can still happen even after sensing, is called the vulnerable time.

- Types of CSMA Access Modes
- There are 4 types of access modes available in CSMA. It is also referred as 4 different types of CSMA protocols which decide the time to start sending data across shared media.
- 1-Persistent: It senses the shared channel first and delivers the data right away if the channel is idle. If not, it must wait and continuously track for the channel to become idle and then broadcast the frame without condition as soon as it does. It is an aggressive transmission algorithm.
- Non-Persistent: It first assesses the channel before transmitting data; if the channel is idle, the node transmits data right away. If not, the station must wait for an arbitrary amount of time (not continuously) and when it discovers the channel is empty, it sends the frames.
- P-Persistent: It consists of the 1-Persistent and Non-Persistent modes combined. Each node observes the channel in the 1-Persistent mode and if the channel is idle, it sends a frame with a P probability. If the data is not transferred, the frame restarts with the following time slot after waiting for a (
  - q = 1 - p
  - q = 1 - p probability) random period.
- O-Persistent: A supervisory node gives each node a transmission order. Nodes wait for their time slot according to their allocated transmission sequence when the transmission medium is idle.

CSMA/CA

- What is CSMA/CA?
- CSMA/CA is a medium access control (MAC) protocol that prevents data collisions in wireless communication.
- It works by avoiding collisions before they occur.
- This is necessary because, in wireless systems, detecting collisions is difficult due to the high energy required for transmission compared to receiving.
- Thus, CSMA/CA forms the backbone of Wi-Fi (IEEE 802.11) and other wireless technologies.

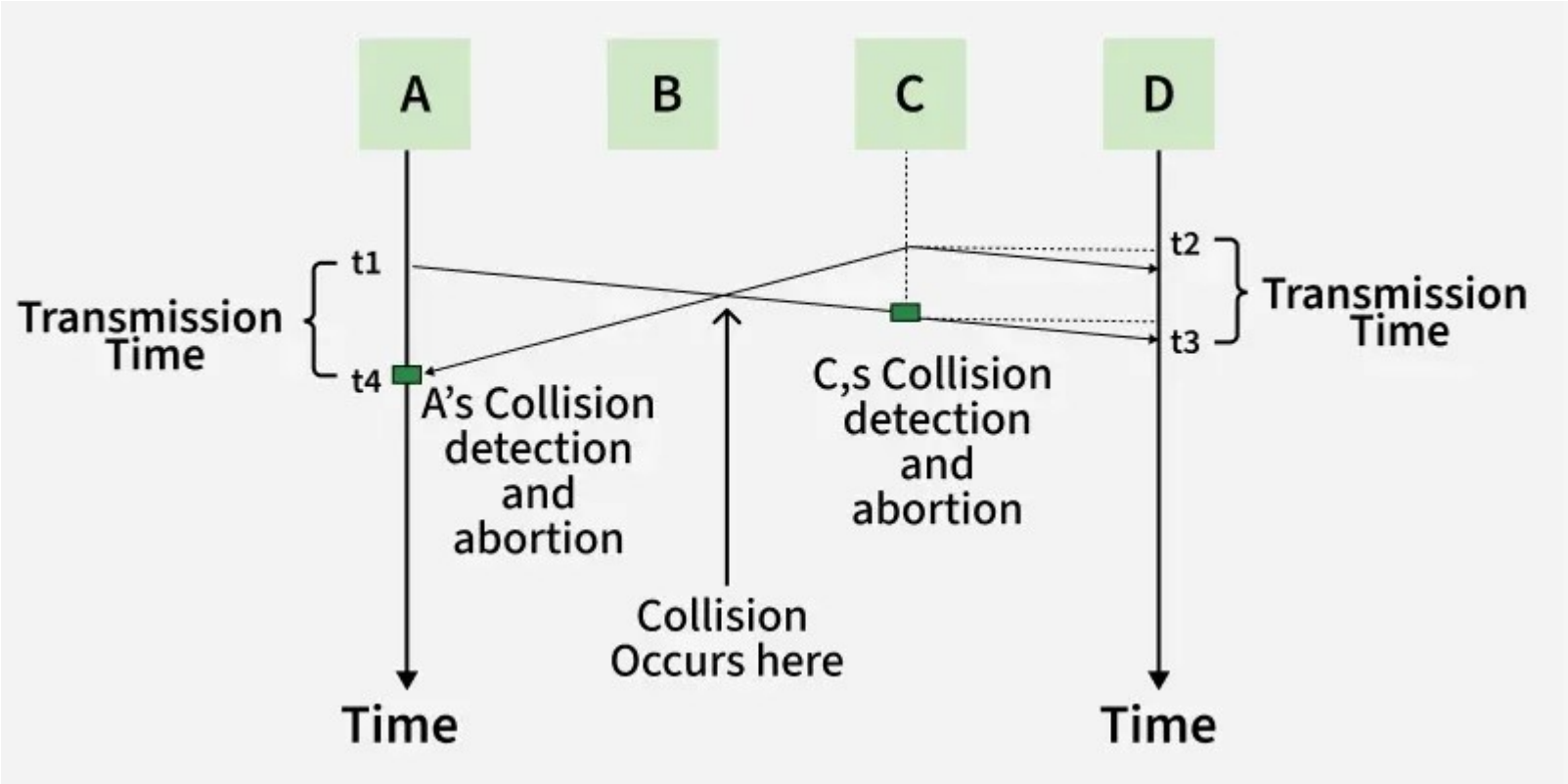
- Why is CSMA/CA Needed?
- In wired networks, devices can detect collisions (CSMA/CD).
- In wireless networks, detecting collisions is nearly impossible because, the transmitting device cannot listen to the channel while sending. Signal interference, noise and hidden terminal problems make detection unreliable.
- Hence, CSMA/CA focuses on collision prevention rather than collision detection.

- How CSMA/CA works
- Carrier Sensing: Before transmitting, a device checks the medium to see if it's busy.
- Collision Avoidance:
- If the channel is idle, the device waits for a random amount of time before sending to avoid multiple devices starting at the same time.
- If the channel is busy, the device waits and senses again later.

- RTS/CTS Handshake: A device sends a "Request to Send" (RTS) message. Other devices hear this and remain silent. The access point then responds with a "Clear to Send" (CTS) message, allowing the original device to transmit.
- Acknowledgments: The receiving device sends an acknowledgment (ACK) to confirm the data was received successfully. If the sender doesn't get an ACK, it assumes a collision occurred and tries again after a longer backoff period (binary exponential backoff).

CSMA/CD

- Collision Detection in CSMA/CD:
- CSMA/CD (Carrier Sense Multiple Access/ Collision Detection) is a media access control method that was widely used in Early Ethernet technology/LANs, when there used to be shared Bus Topology and each node (Computers) was connected by Coaxial Cables.


A
B
C
D
t1
t2
t3
t4
Transmission Time
Transmission Time
A's Collision detection and abortion
C,s Collision detection and abortion
Collision Occurs here
Time
Time

- How Does CSMA/CD Work?
- Step 1: Check if the sender is ready to transmit data packets.
- Step 2:
- Check if the transmission link is idle.
- The sender has to keep on checking if the transmission link/medium is idle.
- For this, it continuously senses transmissions from other nodes.
- The sender sends dummy data on the link. If it does not receive any collision signal, this means the link is idle at the moment.
- If it senses that the carrier is free and there are no collisions, it sends the data. Otherwise, it refrains from sending data.

- Step 3:
- Transmit the data & check for collisions.
- The sender transmits its data on the link.
- CSMA/CD does not use an 'acknowledgment' system. It checks for successful and unsuccessful transmissions through collision signals.
- During transmission, if a collision signal is received by the node, transmission is stopped.
- The station then transmits a jam signal onto the link and waits for random time intervals before it resends the frame.
- After some random time, it again attempts to transfer the data and repeats the above process.
- Step 4: If no collision was detected in propagation, the sender completes its frame transmission and resets the counters.

- Features of CSMA/CD
- Carrier Sense: Before transmitting data, a device listens to the network to check if the transmission medium is free. If the medium is busy, the device waits until it becomes free before transmitting data.
- Multiple Access: In a CSMA/CD network, multiple devices share the same transmission medium. Each device has equal access to the medium and any device can transmit data when the medium is free.

- Collision Detection: If two or more devices transmit data simultaneously, a collision occurs. When a device detects a collision, it immediately stops transmitting and sends a jam signal to inform all other devices on the network of the collision. The devices then wait for a random time before attempting to transmit again, to reduce the chances of another collision.
- Backoff Algorithm: In CSMA/CD, a backoff algorithm is used to determine when a device can retransmit data after a collision. The algorithm uses a random delay before a device retransmits data, to reduce the likelihood of another collision occurring.
- Minimum Frame Size: CSMA/CD requires a minimum frame size to ensure that all devices have enough time to detect a collision before the transmission ends. If a frame is too short, a device may not detect a collision and continue transmitting, leading to data corruption on the network.

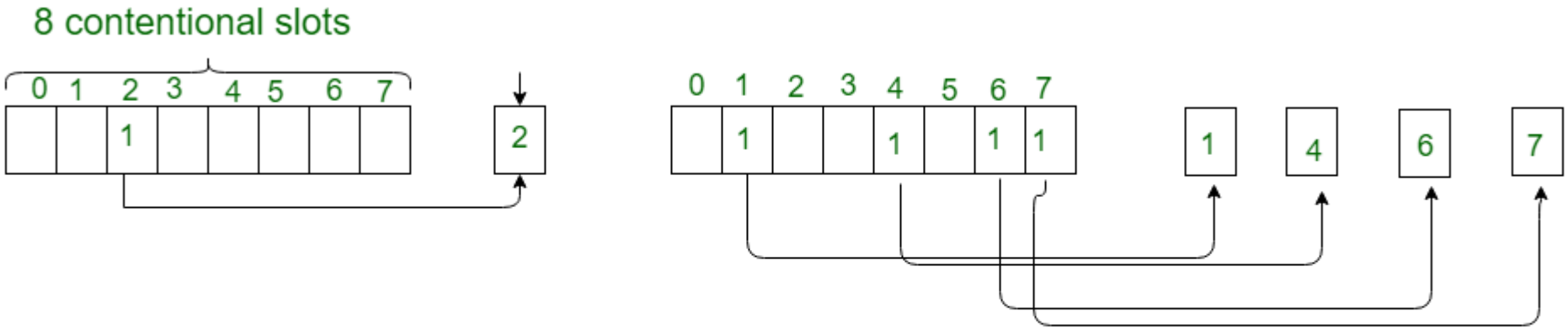
- Limitations of CSMA/CD
- Limited Scalability: CSMA/CD has limitations in terms of scalability and it may not be suitable for large networks with a high number of devices.
- Vulnerability to Collisions: While CSMA/CD can detect collisions, it cannot prevent them from occurring. Collisions can lead to data corruption, retransmission delays and reduced network performance.
- Inefficient Use of Bandwidth: CSMA/CD uses a random backoff algorithm that can result in inefficient use of network bandwidth if a device continually experiences collisions.
- Susceptibility to Security Attacks: CSMA/CD does not provide any security features and the protocol is vulnerable to security attacks such as packet sniffing and spoofing.

Collision Free Protocols

- Almost all collisions can be avoided in CSMA/CD but they can still occur during the contention period. The collision during the contention period adversely affects the system performance, this happens when the cable is long and length of packet are short. This problem becomes serious as fiber optics network came into use. Here we shall discuss some protocols that resolve the collision during the contention period.
- Bit-map Protocol
- Binary Countdown
- Limited Contention Protocols
- The Adaptive Tree Walk Protocol

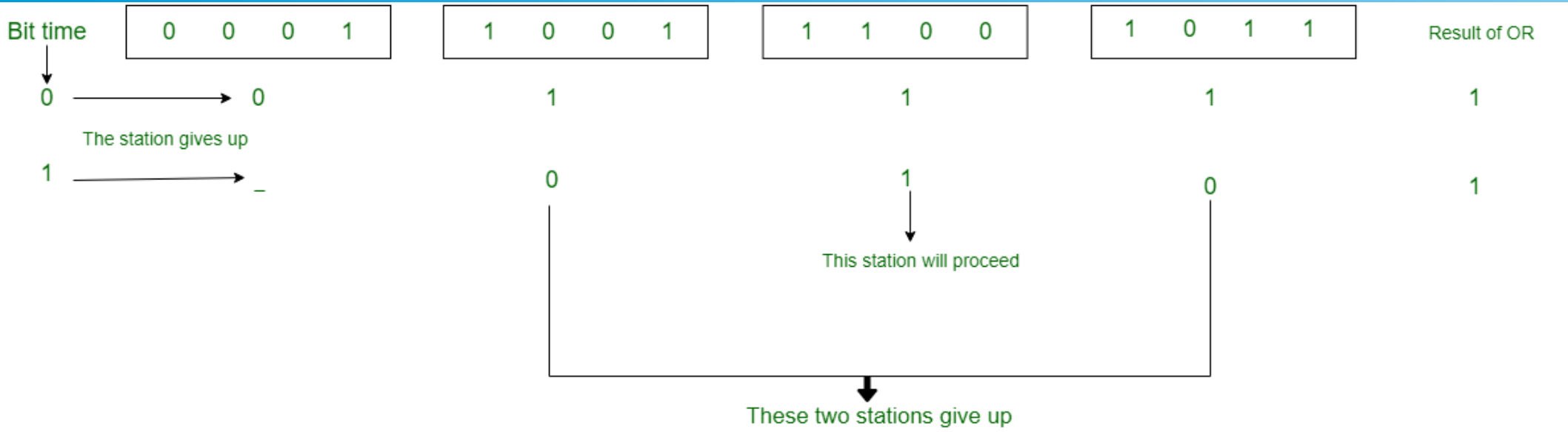
- Pure and slotted Aloha, CSMA and CSMA/CD are Contention based Protocols:
  - Try-if collide-Retry
  - No guarantee of performance
  - What happen if the network load is high?
- Collision Free Protocols:
  - Pay constant overhead to achieve performance guarantee
  - Good when network load is high

- 1. Bit-map Protocol:
- Bit map protocol is collision free Protocol. In bitmap protocol method, each contention period consists of exactly N slots. If any station has to send frame, then it transmits a 1 bit in the corresponding slot. For example, if station 2 has a frame to send, it transmits a 1 bit to the 2nd slot.
- In general, Station 1 Announce the fact that it has a frame questions by inserting a 1 bit into slot 1. In this way, each station has complete knowledge of which station wishes to transmit. There will never be any collisions because everyone agrees on who goes next. Protocols like this in which the desire to transmit is broadcasting for the actual transmission are called Reservation Protocols.



A Bit-map Protocol.

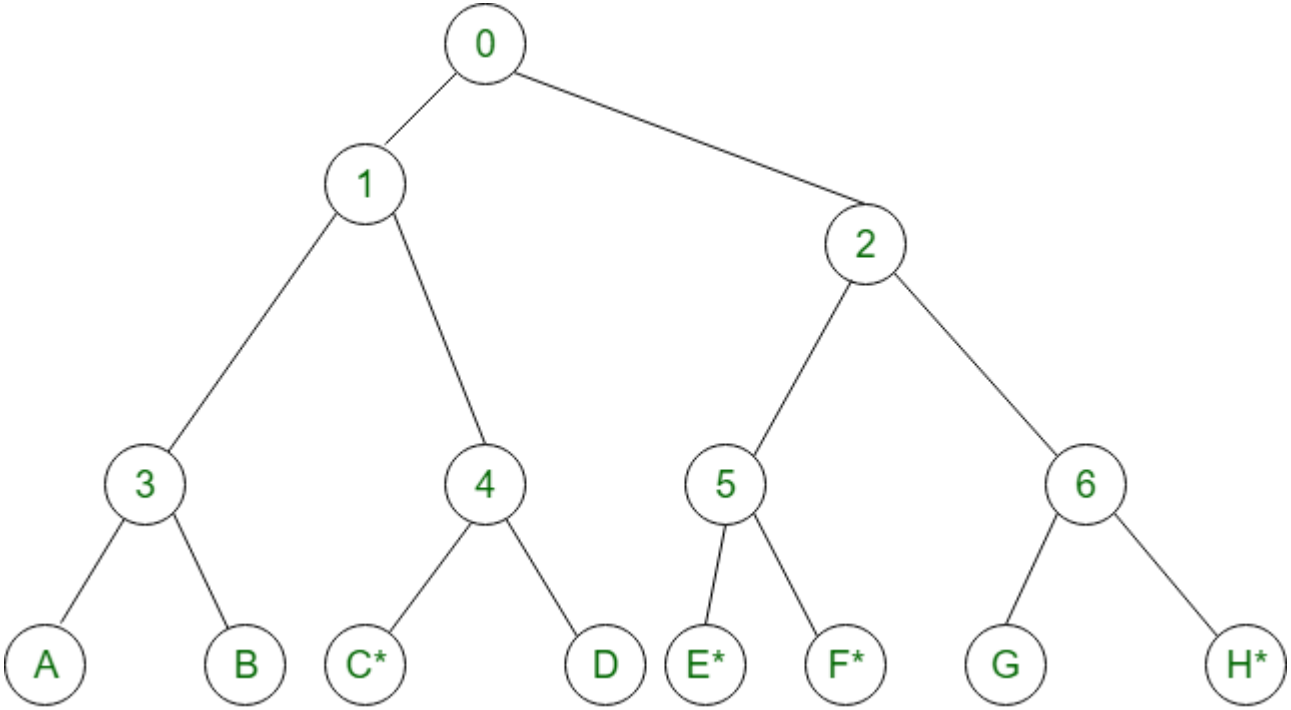
- 2. Binary Countdown:
- Binary countdown protocol is used to overcome the overhead 1 bit per binary station. In binary countdown, binary station addresses are used. A station wanting to use the channel broadcast its address as binary bit string starting with the high order bit. All addresses are assumed of the same length. Here, we will see the example to illustrate the working of the binary countdown.
- In this method, different station addresses are read together who decide the priority of transmitting. If these stations 0001, 1001, 1100, 1011 all are trying to seize the channel for transmission. All the station at first broadcast their most significant address bit that is 0, 1, 1, 1 respectively. The most significant bits are read together. Station 0001 see the 1 MSB in another station address and knows that a higher numbered station is competing for the channel, so it gives up for the current round.
- Other three stations 1001, 1100, 1011 continue. The next station at which next bit is 1 is at station 1100, so station 1011 and 1001 give up because there 2nd bit is 0. Then station 1100 starts transmitting a frame, after which another bidding cycle starts.



Binary countdown

- 3. Limited Contention Protocols:
- Collision based protocols (pure and slotted ALOHA, CSMA/CD) are good when the network load is low.
- Collision free protocols (bitmap, binary Countdown) are good when load is high.
- How about combining their advantages :
- Behave like the ALOHA scheme under light load
- Behave like the bitmap scheme under heavy load.

- 4. Adaptive Tree Walk Protocol:
- partition the group of station and limit the contention for each slot.
- Under light load, everyone can try for each slot like aloha
- Under heavy load, only a group can try for each slot
- How do we do it :
- treat every stations as the leaf of a binary tree
- first slot (after successful transmission), all stations
- can try to get the slot(under the root node).
- If no conflict, fine.
- Else, in case of conflict, only nodes under a subtree get to try for the next one. (depth first search)

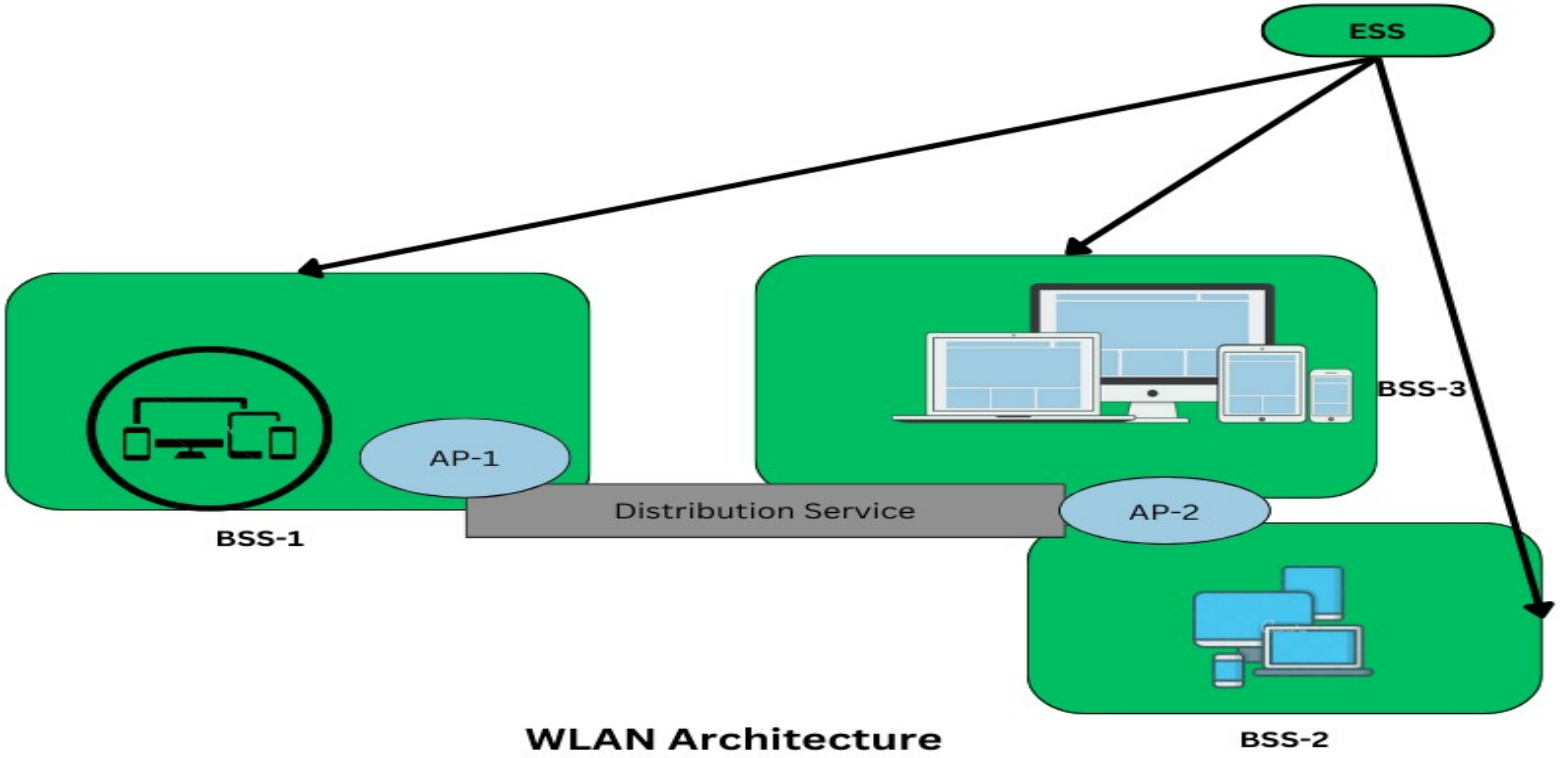

0
1
2
3
4
5
6
A
B
C*
D
E*
F*
G
H*

- Slot-0 : C*, E*, F*, H* (all nodes under node 0 can try which are going to send), conflict
- Slot-1 : C* (all nodes under node 1 can try}, C sends
- Slot-2 : E*, F*, H*(all nodes under node 2 can try}, conflict
- Slot-3 : E*, F* (all nodes under node 5 can try to send), conflict
- Slot-4 : E* (all nodes under E can try), E sends
- Slot-5 : F* (all nodes under F can try), F sends
- Slot-6 : H* (all nodes under node 6 can try to send), H sends.

Wireless LAN Protocol

- WLAN stands for Wireless Local Area Network. WLAN is a local area network that uses radio communication to provide mobility to the network users while maintaining the connectivity to the wired network. A WLAN basically, extends a wired local area network. WLAN's are built by attaching a device called the access point(AP) to the edge of the wired network. Clients communicate with the AP using a wireless network adapter which is similar in function to an ethernet adapter. It is also called a LAWN is a Local area wireless network.

- WLAN Architecture
- Components in Wireless LAN architecture as per IEEE standards are as follows:
- Stations: Stations consist of all the equipment that is used to connect all wireless LANs. Each station has a wireless network controller.
- Base Service Set(BSS): It is a group of stations communicating at the physical layer.
- Extended Service Set(ESS): It is a group of connected Base Service Set(BSS).
- Distribution Service (DS): It connects all Extended Service Set(ESS).



WLAN Architecture

- Types of WLANs
- As per IEEE standard WLAN is categorized into two basic modes, which are as follows:
- Infrastructure: In Infrastructure mode, all the endpoints are connected to a base station and communicate through that; and this can also enable internet access. A WLAN infrastructure can be set up with: a wireless router (base station) and an endpoint (computer, mobile phone, etc). An office or home WiFi connection is an example of Infrastructure mode.
- Ad Hoc: In Ad Hoc mode WLAN connects devices without a base station, like a computer workstation. An Ad Hoc WLAN is easy to set up it provides peer-to-peer communication. It requires two or more endpoints with built-in radio transmission.
- Working of WLAN
- WLAN transmits data over radio signals and the data is sent in the form of a packet. Each packet consists of layers, labels, and instructions with unique MAC addresses assigned to endpoints. This enables routing data packets to correct locations.

- Advantages of WLAN
- Installation speed and simplicity.
- Installation flexibility.
- Reduced cost of ownership.
- Reliability.
- Mobility.
- Robustness.
- Disadvantages of WLAN
- Slower bandwidth.
- Security for wireless LANs is the prime concern.
- Less capacity.
- Wireless networks cost four times more than wired network cards.
- Wireless devices emit low levels of RF which can be harmful to our health.

IEEE
802.11 Architecture and Protocol Stack

- The IEEE 802.11 standard, commonly known as Wi-Fi, outlines the architecture and defines the MAC and physical layer specifications for wireless LANs (WLANs). Wi-Fi uses high-frequency radio waves instead of cables for connecting the devices in LAN. Given the mobility of WLAN nodes, they can move unrestricted within the network coverage zone. The 802.11 structure is designed to accommodate mobile stations that participate actively in network decisions. Furthermore, it can seamlessly integrate with 2G, 3G, and 4G networks.
- The Wi-Fi standard represents a set of wireless LAN standards developed by the Working Group of IEEE LAN/MAN standards committee (IEEE 802). The term 802.11x is also used to denote the set of standards. Various specifications and amendments include 802.11a, 802.11b, 802.11e, 802.11g, 802.11n etc.

- Important Terminologies of IEEE 802.11 Architecture:
- Station: Stations (STA) comprise all devices and equipment that are connected to the wireless LAN. It can be of two types:
- Wireless Access Point (WAP): WAPs or simply access points (AP) are wireless routers that bridge connections for base stations.
- Client: Examples include computers, laptops, printers, and smartphones.
- Access Point: It is a device that can be classified as a station because of its functionalities and acts as a connection between wireless medium and distributed systems.
- Distribution System: A system used to interconnect a set of BSSs and integrated LANs to create an ESS.
- Frame: It is a MAC protocol data unit.

- SSID (Service Set Identifier): It’s the network name for a particular WLAN. All-access points and devices on a specific WLAN must use the same SSID to communicate.
- SDU: It is a data unit that acts as an input to each layer. These can be fragmented or aggregated to form a PDU.
- PDU: It is a data unit projected as an output to communicate with the corresponding layer at the other end. They contain a header specific to the layer.
- Network Interface Controller: It is also known as network interface card. It is a hardware component that connects devices to the network.
- Portal: Serves as a gateway to other networks

802.11
MAC Sublayer Protocol

- MAC layer provides functionality for several tasks like control medium access, can also offer support for roaming, authentication, and power conservation. The basic services provided by MAC are the mandatory asynchronous data service and optional time-bounded service.
- IEEE 802.11 defines two MAC sub-layers:
- Distributed Coordination Function (DCF): DCF uses CSMA/CA as access method as wireless LAN can't implement CSMA/CD. It only offers asynchronous service.
- Point Coordination Function (PCF): PCF is implemented on top of DCF and mostly used for time-service transmission. It uses a centralized, contention-free polling access method. It offers both asynchronous and time-bounded service.

- MAC Frame Structure
- The MAC layer frame structure consists of 9 fields. The following figure shows the basic structure of an IEEE 802.11 MAC data frame along with the content of the frame control field.
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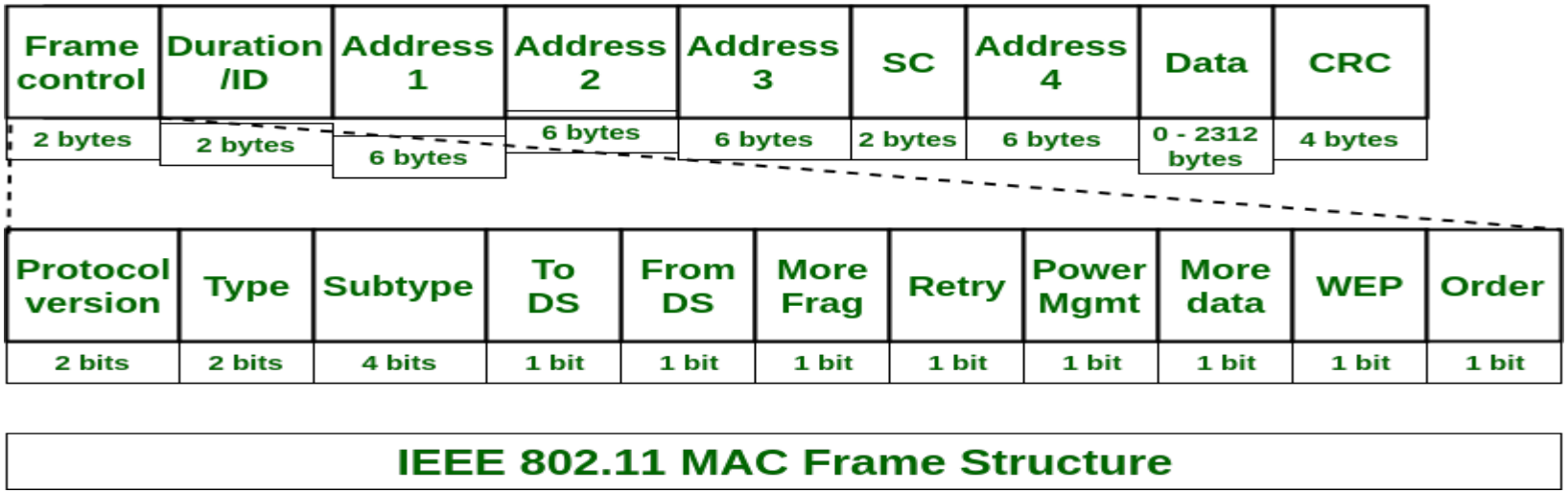


- 1. Frame Control (FC): It is 2 bytes long field which defines type of frame and some control information. Various fields present in FC are:
- Version: It is a 2 bit long field which indicates the current protocol version which is fixed to be 0 for now.
- Type: It is a 2 bit long field which determines the function of frame i.e management(00), control(01) or data(10). The value 11 is reserved.
- Subtype: It is a 4 bit long field which indicates sub-type of the frame like 0000 for association request, 1000 for beacon.
- To DS: It is a 1 bit long field which when set indicates that destination frame is for DS(distribution system).
- From DS: It is a 1 bit long field which when set indicates frame coming from DS.
- More frag (More fragments): It is 1 bit long field which when set to 1 means frame is followed by other fragments.
- Retry: It is 1-bit long field, if the current frame is a retransmission of an earlier frame, this bit is set to 1.

- Power Mgmt (Power management): It is 1-bit long field that indicates the mode of a station after successful transmission of a frame. Set to 1 the field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.
- More data: It is 1-bit long field that is used to indicate receiver that a sender has more data to send than the current frame. This can be used by an access point to indicate to a station in power-save mode that more packets are buffered or it can be used by a station to indicate to an access point after being polled that more polling is necessary as the station has more data ready to transmit.
- WEP: It is 1 bit long field which indicates that the standard security mechanism of 802.11 is applied.
- Order: It is 1 bit long field, if this bit is set to 1 the received frames must be processed in strict order.

- 2. Duration/ID: It is 2 bytes long field which contains the value indicating the period of time in which the medium is occupied(in µs).
- 3. Address 1 to 4: These are 6 bytes long fields which contain standard IEEE 802 MAC addresses (48 bit each). The meaning of each address depends on the DS bits in the frame control field.
- 4. SC (Sequence control): It is 16 bits long field which consists of 2 sub-fields, i.e., Sequence number (12 bits) and Fragment number (4 bits). Since acknowledgement mechanism frames may be duplicated hence, a sequence number is used to filter duplicate frames.
- 5. Data: It is a variable length field which contain information specific to individual frames which is transferred transparently from a sender to the receiver(s).
- 6. CRC (Cyclic redundancy check): It is 4 bytes long field which contains a 32 bit CRC error detection sequence to ensure error free frame.

- Features of the IEEE 802.11 MAC frame
- Frame Control Field: The frame control field contains information about the type of frame, the data rate, and the power management status.
- Duration Field: The duration field specifies the length of time that the channel will be occupied by the transmission.
- Address Fields: The address fields specify the source and destination MAC addresses of the Wi-Fi devices involved in the communication.
- Sequence Control Field: The sequence control field is used to identify and manage the transmission sequence of the frames.
- Frame Body: The frame body contains the actual data being transmitted between Wi-Fi devices, such as IP packets, TCP segments, or UDP datagrams.
- Frame Check Sequence: The frame check sequence (FCS) is used to check the integrity of the data transmitted in the frame and to detect any transmission errors.
- Management, Control, and Data Frames: The IEEE 802.11 MAC frame defines three types of frames: management frames, control frames, and data frames. Management frames are used for network management, control frames are used for coordination between Wi-Fi devices, and data frames are used for the transmission of actual data.
- Fragmentation: The IEEE 802.11 MAC frame supports fragmentation, which allows large data packets to be divided into smaller fragments for transmission.
- Acknowledgments: The IEEE 802.11 MAC frame uses acknowledgments to confirm the successful transmission of frames and to request the retransmission of any frames that were not successfully received.

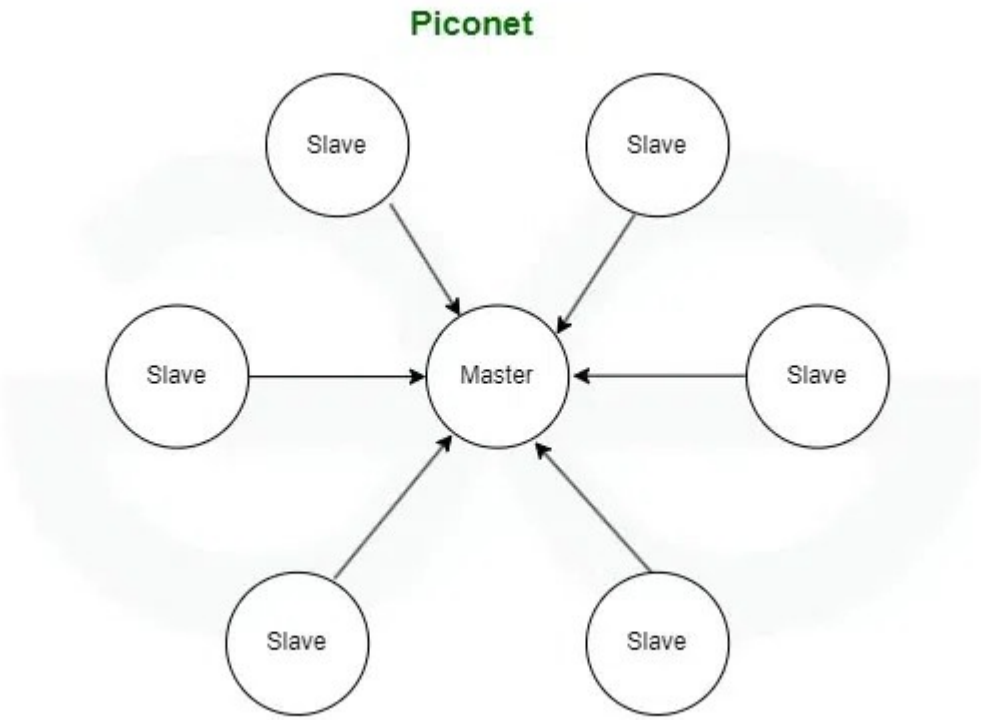
Bluetooth Architecture

- Bluetooth is used for short-range wireless voice and data communication. It is a Wireless Personal Area Network (WPAN) technology and is used for data communications over smaller distances. This generation changed into being invented via Ericson in 1994. It operates within the unlicensed, business, scientific, and clinical (ISM) bands from 2.4 GHz to 2.485 GHz.
- Bluetooth stages up to 10 meters. Depending upon the version, it presents information up to at least 1 Mbps or 3 Mbps. The spreading method that it uses is FHSS (Frequency-hopping unfold spectrum). A Bluetooth network is called a piconet and a group of interconnected piconets is called a scatter net.

- Bluetooth is a wireless technology that lets devices like phones, tablets, and headphones connect to each other and share information without needing cables. Bluetooth simply follows the principle of transmitting and receiving data using radio waves. It can be paired with the other device which has also Bluetooth but it should be within the estimated communication range to connect. When two devices start to share data, they form a network called piconet which can further accommodate more than five devices.
- Key Features of Bluetooth
- The transmission capacity of Bluetooth is 720 kbps.
- Bluetooth is a wireless technology.
- Bluetooth is a Low-cost and short-distance radio communications standard.
- Bluetooth is robust and flexible.
- The basic architecture unit of Bluetooth is a piconet.

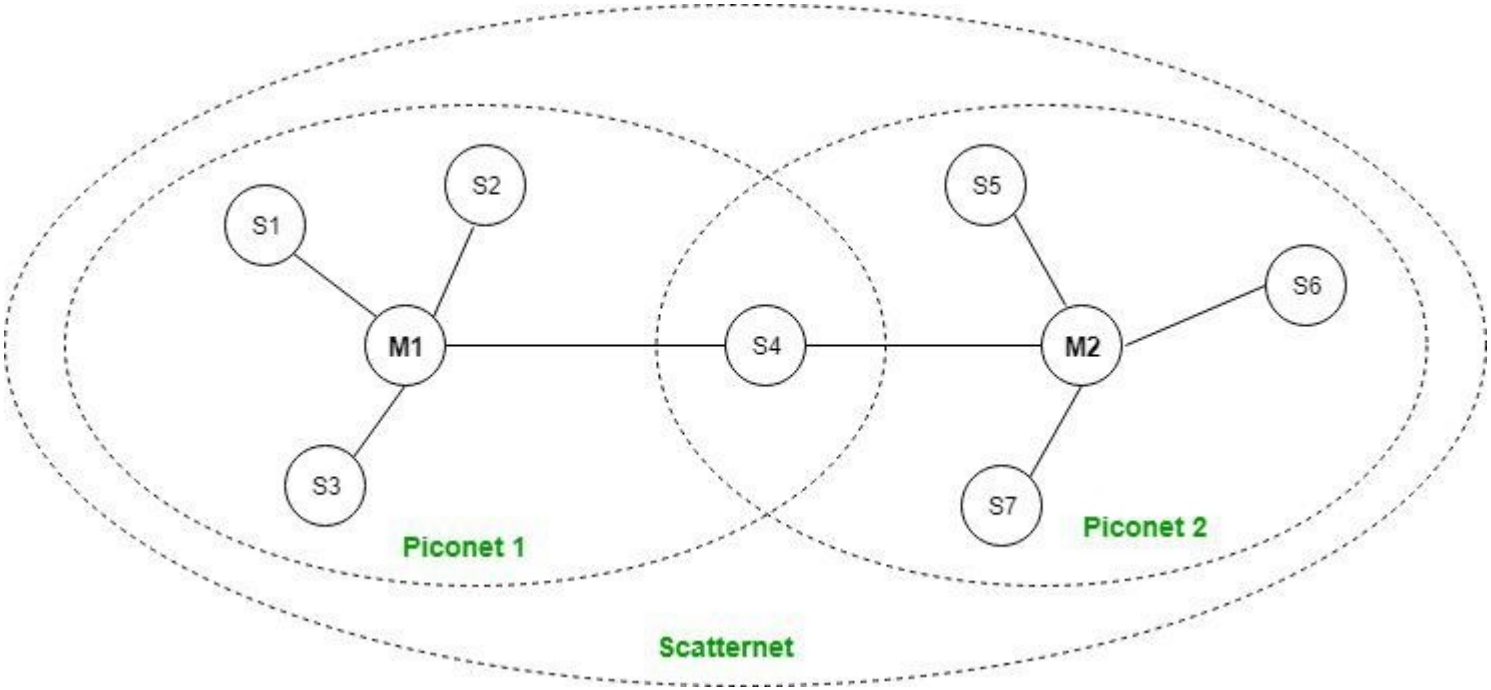
Architecture of Bluetooth

- Piconet
- Piconet is a type of Bluetooth network that contains one primary node called the master node and seven active secondary nodes called slave nodes. Thus, we can say that there is a total of 8 active nodes which are present at a distance of 10 meters. The communication between the primary and secondary nodes can be one-to-one or one-to-many. Possible communication is only between the master and slave; Slave-slave communication is not possible. It also has 255 parked nodes, these are secondary nodes and cannot take participation in communication unless it gets converted to the active state.

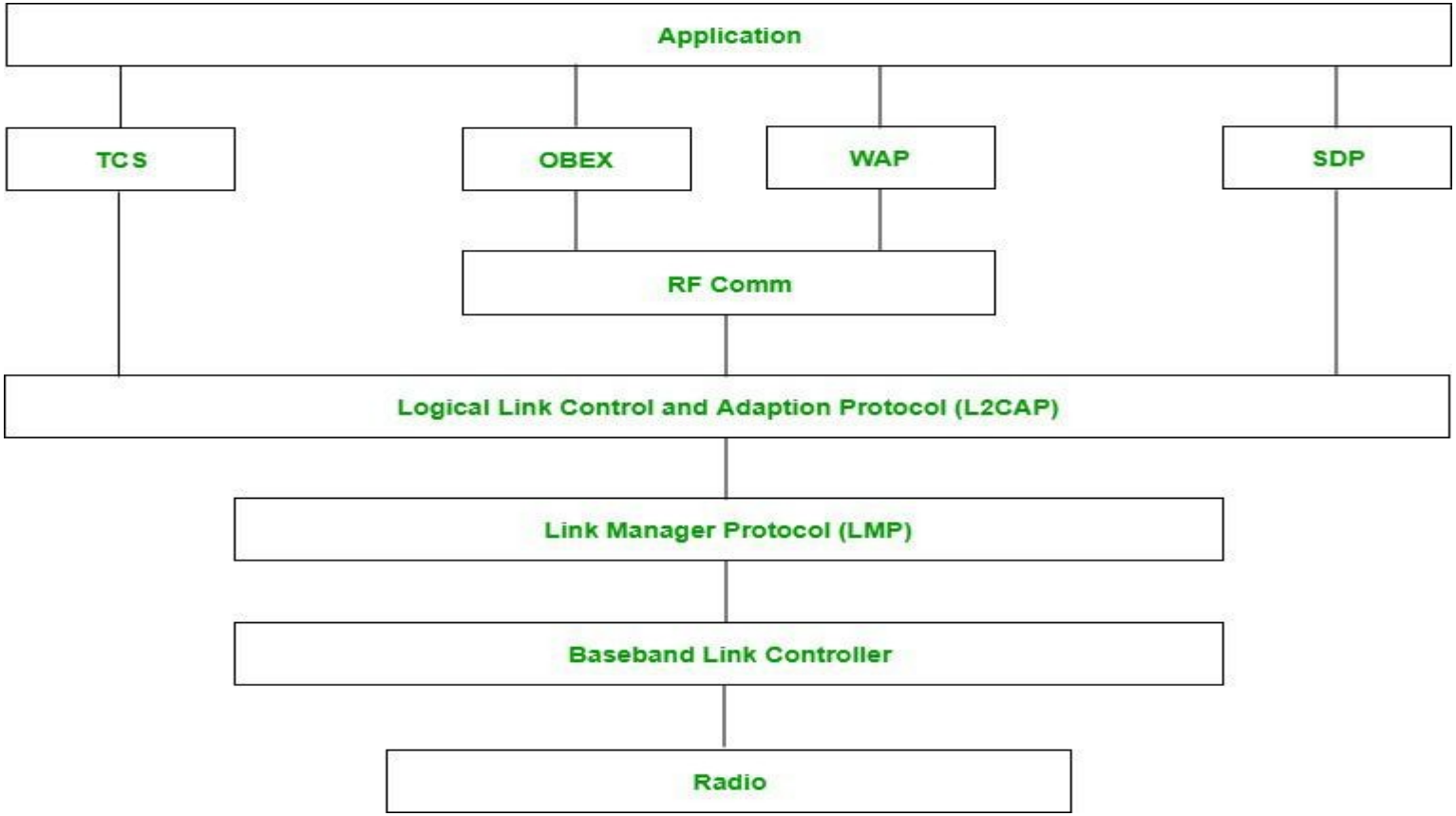


What is Piconet ?

- Scatternet
- It is formed by using various piconets. A slave that is present in one piconet can act as master or we can say primary in another piconet. This kind of node can receive a message from a master in one piconet and deliver the message to its slave in the other piconet where it is acting as a master. This type of node is referred to as a bridge node. A station cannot be mastered in two piconets.


S1
S2
M1
S3
S4
S5
M2
S6
S7
Piconet 1
Piconet 2
Scatternet

- Bluetooth Protocol Stack
- Radio (RF) Layer: It specifies the details of the air interface, including frequency, the use of frequency hopping and transmit power. It performs modulation/demodulation of the data into RF signals. It defines the physical characteristics of Bluetooth transceivers. It defines two types of physical links: connection-less and connection-oriented.
- Baseband Link Layer: The baseband is the digital engine of a Bluetooth system and is equivalent to the MAC sublayer in LANs. It performs the connection establishment within a piconet, addressing, packet format, timing and power control.
- Link Manager Protocol Layer: It performs the management of the already established links which includes authentication and encryption processes. It is responsible for creating the links, monitoring their health, and terminating them gracefully upon command or failure.
- Logical Link Control and Adaption (L2CAP) Protocol Layer: It is also known as the heart of the Bluetooth protocol stack. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs segmentation and multiplexing.
- Service Discovery Protocol (SDP) Layer: It is short for Service Discovery Protocol. It allows discovering the services available on another Bluetooth-enabled device.
- RF Comm Layer: It is a cabal replacement protocol. It is short for Radio Frontend Component. It provides a serial interface with WAP and OBEX. It also provides emulation of serial ports over the logical link control and adaption protocol(L2CAP). The protocol is based on the ETSI standard TS 07.10.
- OBEX: It is short for Object Exchange. It is a communication protocol to exchange objects between 2 devices.
- WAP: It is short for Wireless Access Protocol. It is used for internet access.
- TCS: It is short for Telephony Control Protocol. It provides telephony service. The basic function of this layer is call control (setup & release) and group management for the gateway serving multiple devices.
- Application Layer: It enables the user to interact with the application.


Application
TCS
OBEX
WAP
SDP
RF Comm
Logical Link Control and Adaption Protocol (L2CAP)
Link Manager Protocol (LMP)
Baseband Link Controller
Radio

- Types of Bluetooth
- Various types of Bluetooth are available in the market nowadays. Let us look at them.
- In-Car Headset: One can make calls from the car speaker system without the use of mobile phones.
- Stereo Headset: To listen to music in car or in music players at home.
- Webcam: One can link the camera with the help of Bluetooth with their laptop or phone.
- Bluetooth-Equipped Printer: The printer can be used when connected via Bluetooth with mobile phone or laptop.
- Bluetooth Global Positioning System (GPS): To use Global Positioning System (GPS) in cars, one can connect their phone with car system via Bluetooth to fetch the directions of the address.

- Applications of Bluetooth
- It can be used in wireless headsets, wireless PANs, and LANs.
- It can connect a digital camera wireless to a mobile phone.
- It can transfer data in terms of videos, songs, photographs, or files from one cell phone to another cell phone or computer.
- It is used in the sectors of Medical healthcare, sports and fitness, Military.
- Advantages
- It is a low-cost and easy-to-use device.
- It can also penetrate through walls.
- It creates an Ad-hoc connection immediately without any wires.
- It is used for voice and data transfer.
- Disadvantages
- It can be hacked and hence, less secure.
- It has a slow data transfer rate of 3 Mbps.
- Bluetooth communication does not support routing.

Thank You